

Table 1: End-to-end execution time (s, median per iteration) and speedup split by pass type. ADT fields = non-recursive fields annotated with `@BENCH adt_fields`; SoA bufs = 1+ non-recursive field slots across all constructors (recursive children are stored in the tag buffer). Speedup $>1\times$ means SoA is faster; **bold** marks $>1.1\times$.

Program	ADT fields	SoA bufs	Fold passes			Map passes		
			AoS (s)	SoA (s)	Speedup	AoS (s)	SoA (s)	Speedup
Compiler	9	8	0.1439	0.0668	2.15 $\times$	0.1606	0.1776	0.90 $\times$
DBQuery	15	13	0.1323	0.0962	1.38 $\times$	0.1373	0.1671	0.82 $\times$
DecisionTree	7	6	3.981	2.929	1.36 $\times$	—	—	—
DomTree	15	14	0.1362	0.0780	1.75 $\times$	0.0588	0.0809	0.73 $\times$
KDTree	17	16	0.1762	0.1432	1.23 $\times$	—	—	—
LinearListReduction	11	11	0.0303	0.0065	4.63 $\times$	—	—	—
List	2	2	0.1301	0.1119	1.16 $\times$	0.2556	0.3424	0.75 $\times$
MonoTree	3	2	0.0175	0.0223	0.78 $\times$	0.0945	0.0752	1.26 $\times$
ObjectGraph	5	4	0.0890	0.0927	0.96 $\times$	0.1427	0.1615	0.88 $\times$
OctTree_sumMass	16	—	0.0352	0.0300	1.17 $\times$	—	—	—
OctTree_sumEnergy	16	—	0.0569	0.0349	1.63 $\times$	—	—	—
OctTree_countActive	16	—	0.0426	0.0342	1.25 $\times$	—	—	—
OctTree_countParticles	16	—	0.0411	0.0381	1.08 $\times$	—	—	—
OctTree_barnesHutPotential	16	—	0.0467	0.0580	0.80 $\times$	—	—	—
OctTree_fmmPotential	16	—	0.1087	0.1015	1.07 $\times$	—	—	—
OctTree_scaleEnergy	16	—	—	—	—	0.1552	0.1769	0.88 $\times$
OctTree_clearFlags	16	—	—	—	—	0.1547	0.1760	0.88 $\times$
PiecewiseFunctions	8	7	0.1356	0.0958	1.42 $\times$	0.2385	0.2615	0.91 $\times$
TernaryTree	5	3	0.0288	0.0256	1.13 $\times$	0.0835	0.0915	0.91 $\times$
Trie	10	9	0.0632	0.0628	1.01 $\times$	0.1508	0.1665	0.91 $\times$
ColorOctree	25	18	0.0901	0.0794	1.13 $\times$	—	—	—

Table 2: PAPI speedup summary across all passes. For each program and counter, speedup is AoS/SoA using summed per-pass median counters. $>1\times$ means SoA has fewer events.

Program	CYC	INS	L1D	L1I	L2D	L2I	LLC
Compiler	1.50 $\times$	0.52 $\times$	2.75 $\times$	0.86 $\times$	10.34 $\times$	0.36 $\times$	7.34 $\times$
DBQuery	1.03 $\times$	0.47 $\times$	1.46 $\times$	0.46 $\times$	8.55 $\times$	0.31 $\times$	5.90 $\times$
DecisionTree	1.31 $\times$	0.59 $\times$	17.51 $\times$	5.03 $\times$	0.90 $\times$	4.45 $\times$	36.42 $\times$
DomTree	1.51 $\times$	0.57 $\times$	2.01 $\times$	0.82 $\times$	19.18 $\times$	0.21 $\times$	11.47 $\times$
KDTree	1.26 $\times$	0.77 $\times$	1.28 $\times$	1.13 $\times$	0.24 $\times$	1.05 $\times$	4.51 $\times$
LinearListReduction	5.43 $\times$	0.90 $\times$	1.23 $\times$	2.50 $\times$	4.46 $\times$	3.62 $\times$	11.15 $\times$
List	0.80 $\times$	0.80 $\times$	1.79 $\times$	0.47 $\times$	0.09 $\times$	0.54 $\times$	1.47 $\times$
MonoTree	0.98 $\times$	0.85 $\times$	1.32 $\times$	0.70 $\times$	0.90 $\times$	0.74 $\times$	1.25 $\times$
ObjectGraph	0.95 $\times$	0.75 $\times$	4.76 $\times$	0.85 $\times$	0.45 $\times$	0.79 $\times$	2.90 $\times$
OctTree_sumMass	1.16 $\times$	0.89 $\times$	0.62 $\times$	1.29 $\times$	0.10 $\times$	1.46 $\times$	5.00 $\times$
OctTree_sumEnergy	0.92 $\times$	0.87 $\times$	0.17 $\times$	1.30 $\times$	0.03 $\times$	1.37 $\times$	2.19 $\times$
OctTree_countActive	1.26 $\times$	1.01 $\times$	7.99 $\times$	1.83 $\times$	12.48 $\times$	2.33 $\times$	20.35 $\times$
OctTree_countParticles	1.33 $\times$	1.00 $\times$	35.94 $\times$	1.93 $\times$	35.58 $\times$	2.98 $\times$	42.16 $\times$
OctTree_barnesHutPotential	0.84 $\times$	0.82 $\times$	3.96 $\times$	0.12 $\times$	4.19 $\times$	0.31 $\times$	2.92 $\times$
OctTree_fmmPotential	1.05 $\times$	0.87 $\times$	1.55 $\times$	0.68 $\times$	1.39 $\times$	0.60 $\times$	2.19 $\times$
OctTree_scaleEnergy	0.81 $\times$	0.49 $\times$	0.56 $\times$	0.43 $\times$	0.25 $\times$	0.64 $\times$	2.29 $\times$
OctTree_clearFlags	0.83 $\times$	0.46 $\times$	0.45 $\times$	0.54 $\times$	0.41 $\times$	0.52 $\times$	2.34 $\times$
PiecewiseFunctions	1.11 $\times$	0.60 $\times$	2.87 $\times$	0.61 $\times$	2.02 $\times$	0.50 $\times$	4.44 $\times$
TernaryTree	1.16 $\times$	0.79 $\times$	0.33 $\times$	0.54 $\times$	0.23 $\times$	1.11 $\times$	1.43 $\times$
Trie	1.14 $\times$	0.56 $\times$	1.83 $\times$	0.82 $\times$	1.98 $\times$	0.20 $\times$	6.68 $\times$
ColorOctree	1.13 $\times$	0.84 $\times$	4.14 $\times$	0.88 $\times$	0.71 $\times$	0.79 $\times$	4.40 $\times$
Geomean	1.17 $\times$	0.71 $\times$	1.92 $\times$	0.85 $\times$	1.19 $\times$	0.80 $\times$	4.88 $\times$

Table 3: PAPI counter totals across all passes (AoS/SoA). Each entry is the sum of per-pass median counter values.

Program	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
Compiler	8.8e+08/ 5.8e+08	9.9e+08 /1.9e+09	1.3e+07/ 4.9e+06	2.9e+05 /3.3e+05	1.4e+06/ 1.4e+05	1.1e+04 /3.1e+04	5.1e+07/ 6.9e+06
DBQuery	9.2e+08/ 8.9e+08	1.1e+09 /2.2e+09	5.2e+06/ 3.5e+06	1.6e+05 /3.5e+05	8.8e+05/ 1.0e+05	8.4e+03 /2.7e+04	2.1e+07/ 3.5e+06
DecisionTree	2.0e+10/ 1.5e+10	4.0e+10 /6.8e+10	5.4e+08/ 3.1e+07	1.3e+05/ 2.5e+04	1.9e+06 /2.1e+06	1.1e+05/ 2.4e+04	1.1e+09/ 3.0e+07
DomTree	8.2e+08/ 5.4e+08	1.2e+09 /2.2e+09	1.1e+07/ 5.7e+06	1.0e+05 /1.2e+05	1.1e+06/ 5.8e+04	6.6e+03 /3.2e+04	5.7e+07/ 5.0e+06
KDTree	8.9e+08/ 7.1e+08	3.1e+09 /4.0e+09	1.4e+07/ 1.1e+07	2.1e+03/ 1.8e+03	2.6e+04 /1.1e+05	1.7e+03/ 1.7e+03	4.4e+07/ 9.8e+06
LinearListReduction	1.4e+08/ 2.5e+07	9.0e+07 /1.0e+08	1.5e+06/ 1.2e+06	7.4e+02/ 3.0e+02	3.0e+05/ 6.8e+04	6.4e+02/ 1.8e+02	1.2e+07/ 1.1e+06
List	1.3e+09 /1.6e+09	4.5e+09 /5.6e+09	9.4e+06/ 5.2e+06	2.6e+05 /5.5e+05	2.1e+04 /2.3e+05	7.0e+03 /1.3e+04	4.9e+07/ 3.3e+07
MonoTree	2.5e+08 /2.6e+08	9.1e+08 /1.1e+09	7.4e+05/ 5.6e+05	3.3e+04 /4.8e+04	1.3e+04 /1.5e+04	2.3e+03 /3.1e+03	2.2e+06/ 1.8e+06
ObjectGraph	8.3e+08 /8.7e+08	3.0e+09 /4.0e+09	1.0e+07/ 2.1e+06	2.0e+05 /2.4e+05	3.0e+04 /6.7e+04	1.1e+04 /1.4e+04	2.4e+07/ 8.4e+06
OctTree_sumMass	1.7e+08/ 1.5e+08	5.7e+08 /6.3e+08	8.5e+05 /1.4e+06	5.5e+02/ 4.3e+02	2.8e+03 /2.9e+04	4.0e+02/ 2.8e+02	6.2e+06/ 1.2e+06
OctTree_sumEnergy	1.6e+08 /1.7e+08	7.7e+08 /8.9e+08	4.7e+05 /2.7e+06	5.6e+02/ 4.3e+02	2.1e+03 /6.0e+04	4.1e+02/ 3.0e+02	5.3e+06/ 2.4e+06
OctTree_countActive	2.1e+08/ 1.6e+08	5.8e+08/ 5.8e+08	7.3e+06/ 9.1e+07	7.8e+02 / 4.2e+02	2.9e+05/ 2.3e+04	5.9e+02/ 2.5e+02	6.1e+06/ 3.0e+05
OctTree_countParticles	2.0e+08/ 1.5e+08	5.7e+08/ 5.7e+08	7.3e+06/ 2.0e+05	8.0e+02/ 4.1e+02	2.9e+05/ 8.3e+03	6.2e+02/ 2.1e+02	6.1e+06/ 1.4e+05
OctTree_barnesHutPotential	2.3e+08 /2.7e+08	8.2e+08 /1.0e+09	4.9e+06/ 1.2e+06	4.6e+02 /3.7e+03	1.5e+05/ 3.7e+04	3.4e+02 /1.1e+03	5.8e+06/ 2.0e+06
OctTree_fmmPotential	5.3e+08/ 5.0e+08	1.6e+09 /1.8e+09	7.9e+06/ 5.1e+06	7.4e+02 /1.1e+03	1.4e+05/ 1.0e+05	5.7e+02 /9.4e+02	4.6e+06/ 2.1e+06
OctTree_scaleEnergy	3.6e+08 /4.4e+08	9.1e+08 /1.9e+09	1.4e+06 /2.5e+06	1.7e+05 /3.9e+05	9.8e+03 /3.9e+04	7.2e+03 /1.1e+04	5.7e+06/ 2.5e+06
OctTree_clearFlags	3.7e+08 /4.4e+08	8.0e+08 /1.7e+09	1.0e+06 /2.3e+06	2.2e+05 /4.0e+05	1.5e+04 /3.6e+04	4.9e+03 /9.5e+03	5.6e+06/ 2.4e+06
PiecewiseFunctions	1.2e+09/ 1.1e+09	2.8e+09 /4.7e+09	1.4e+07/ 5.0e+06	2.8e+05 /4.5e+05	2.0e+05/ 9.7e+04	2.1e+04 /4.3e+04	4.5e+07/ 1.0e+07
TernaryTree	4.2e+08/ 3.6e+08	1.2e+09 /1.5e+09	4.6e+05 /1.4e+06	6.7e+04 /1.2e+05	4.6e+03 /2.0e+04	3.1e+03/ 2.8e+03	3.3e+06/ 2.3e+06
Trie	6.5e+08/ 5.7e+08	1.2e+09 /2.2e+09	8.4e+06/ 4.6e+06	1.8e+05 /2.2e+05	1.6e+05/ 8.2e+04	6.6e+03 /3.3e+04	2.6e+07/ 3.9e+06
ColorOctree	4.4e+08/ 4.0e+08	1.6e+09 /2.0e+09	9.9e+06/ 2.4e+06	7.3e+02 /8.2e+02	6.1e+04 /8.5e+04	5.2e+02 /6.5e+02	1.9e+07/ 4.3e+06

Table 4: Mutable vs immutable cursor comparison (times only). Times are median per iteration (s). **Bold** marks the fastest time across all four variants.

Program	Am	Ai	Sm	Si
Compiler	0.3045	1.112	0.2444	<i>OOM</i>
DBQuery	0.2696	0.3441	0.2633	1.119
DecisionTree	3.981	5.823	2.929	24.95
DomTree	0.1950	0.3350	0.1589	0.6521
KDTree	0.1762	0.2505	0.1432	0.7698
LinearListReduction	0.0303	0.1396	0.0065	<i>OOM</i>
List	0.3857	<i>OOM</i>	0.4542	<i>OOM</i>
MonoTree	0.1120	0.1934	0.0975	0.2844
ObjectGraph	0.2317	0.5018	0.2542	0.7045
OctTree_sumMass	0.0352	0.0459	0.0300	<i>OOM</i>
OctTree_sumEnergy	0.0569	0.0599	0.0349	<i>OOM</i>
OctTree_countActive	0.0426	0.0572	0.0342	<i>OOM</i>
OctTree_countParticles	0.0411	0.0561	0.0381	<i>OOM</i>
OctTree_barnesHutPotential	0.0467	0.0704	0.0580	<i>OOM</i>
OctTree_fmmPotential	0.1087	0.0999	0.1015	<i>OOM</i>
OctTree_scaleEnergy	0.1552	0.1547	0.1769	<i>OOM</i>
OctTree_clearFlags	0.1547	0.2017	0.1760	<i>OOM</i>
PiecewiseFunctions	0.3741	0.5699	0.3574	2.169
TernaryTree	0.1124	0.1523	0.1171	0.6183
Trie	0.2140	0.3334	0.2293	1.022
ColorOctree	0.0901	0.1219	0.0794	<i>OOM</i>

Table 5: Mutable vs immutable cursor comparison (speedups). Shown: AoS-mut/SoA-mut, AoS-imm/AoS-mut, AoS-imm/SoA-mut, AoS-imm/SoA-imm. $>1\times$ means the denominator is faster.

Program	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
Compiler	1.25 $\times$	3.65 $\times$	4.55 $\times$	–
DBQuery	1.02 $\times$	1.28 $\times$	1.31 $\times$	0.31 $\times$
DecisionTree	1.36 $\times$	1.46 $\times$	1.99 $\times$	0.23 $\times$
DomTree	1.23 $\times$	1.72 $\times$	2.11 $\times$	0.51 $\times$
KDTree	1.23 $\times$	1.42 $\times$	1.75 $\times$	0.33 $\times$
LinearListReduction	4.63 $\times$	4.60 $\times$	21.33 $\times$	–
List	0.85 $\times$	–	–	–
MonoTree	1.15 $\times$	1.73 $\times$	1.98 $\times$	0.68 $\times$
ObjectGraph	0.91 $\times$	2.17 $\times$	1.97 $\times$	0.71 $\times$
OctTree_sumMass	1.17 $\times$	1.30 $\times$	1.53 $\times$	–
OctTree_sumEnergy	1.63 $\times$	1.05 $\times$	1.72 $\times$	–
OctTree_countActive	1.25 $\times$	1.34 $\times$	1.67 $\times$	–
OctTree_countParticles	1.08 $\times$	1.37 $\times$	1.47 $\times$	–
OctTree_barnesHutPotential	0.80 $\times$	1.51 $\times$	1.21 $\times$	–
OctTree_fmmPotential	1.07 $\times$	0.92 $\times$	0.98 $\times$	–
OctTree_scaleEnergy	0.88 $\times$	1.00 $\times$	0.87 $\times$	–
OctTree_clearFlags	0.88 $\times$	1.30 $\times$	1.15 $\times$	–
PiecewiseFunctions	1.05 $\times$	1.52 $\times$	1.59 $\times$	0.26 $\times$
TernaryTree	0.96 $\times$	1.36 $\times$	1.30 $\times$	0.25 $\times$
Trie	0.93 $\times$	1.56 $\times$	1.45 $\times$	0.33 $\times$
ColorOctree	1.13 $\times$	1.35 $\times$	1.54 $\times$	–

Table 6: Per-pass performance for **Compiler**, ADT has 9 fields, SoA uses 8 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across $-iterate$ runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
blockCountPass	F	2/9	78%	0.0126	0.0800	0.0037	<i>OOM</i>	3.44 $\times$	6.35 $\times$	21.87 $\times$	–
branchStatsPass	F	2/9	78%	0.0114	0.0803	0.0046	<i>OOM</i>	2.48 $\times$	7.04 $\times$	17.44 $\times$	–
castInstCountPass	F	3/9	67%	0.0179	0.0803	0.0021	<i>OOM</i>	8.69 $\times$	4.49 $\times$	39.00 $\times$	–
has cycle	F	4/9	56%	0.0344	0.0934	0.0324	<i>OOM</i>	1.06 $\times$	2.71 $\times$	2.88 $\times$	–
instCountPass	F	2/9	78%	0.0144	0.0799	0.0042	<i>OOM</i>	3.41 $\times$	5.56 $\times$	18.96 $\times$	–
latencyModelPass	F	3/9	67%	0.0116	0.0844	0.0034	<i>OOM</i>	3.38 $\times$	7.26 $\times$	24.51 $\times$	–
memoryOpStatsPass	F	3/9	67%	0.0116	0.0803	0.0064	<i>OOM</i>	1.82 $\times$	6.94 $\times$	12.61 $\times$	–
stripSideEffectsPass	M	7/9	22%	0.0805	0.1816	0.0903	<i>OOM</i>	0.89 $\times$	2.26 $\times$	2.01 $\times$	–
targetReturnPass	M	9/9	0%	0.0801	0.2010	0.0873	<i>OOM</i>	0.92 $\times$	2.51 $\times$	2.30 $\times$	–
throughputModelPass	F	3/9	67%	0.0118	0.0806	0.0030	<i>OOM</i>	3.96 $\times$	6.84 $\times$	27.07 $\times$	–
verifyIR	F	9/9	0%	0.0183	0.0705	0.0071	<i>OOM</i>	2.58 $\times$	3.86 $\times$	9.97 $\times$	–
Total				0.3045	1.112	0.2444	–	1.25 $\times$	3.65 $\times$	4.55 $\times$	–
Geomean								2.38 $\times$	4.67 $\times$	11.10 $\times$	–

Table 7: Per-pass PAPI counters for **Compiler** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
blockCountPass	F	4.8e+07/ 9.9e+06	5.1e+07/ 4.5e+07	1.3e+06/ 1.0e+05	3.4e+02/ 8.3e+01	2.5e+05/ 2.7e+03	3.0e+02/ 5.9e+01	4.7e+06/ 1.3e+04
branchStatsPass	F	5.8e+07/ 1.6e+07	7.9e+07 /8.6e+07	8.1e+05/ 2.2e+04	2.3e+02/ 1.4e+02	4.9e+04/ 3.4e+03	1.9e+02/ 1.3e+02	4.7e+06/ 4.8e+05
castInstCountPass	F	9.1e+07/ 6.6e+06	5.1e+07/ 3.9e+07	1.4e+04 /3.3e+04	1.9e+02/ 8.7e+01	8.4e+02 /1.1e+03	1.9e+02/ 6.3e+01	4.7e+06/ 1.7e+04
has cycle	F	1.3e+08/ 8.8e+07	1.1e+08 /1.5e+08	2.4e+06/ 5.8e+05	9.0e+04/ 5.9e+04	1.1e+05/ 2.2e+04	2.8e+03/ 5.2e+02	5.0e+06/ 1.5e+06
instCountPass	F	3.9e+07/ 1.1e+07	5.6e+07 /5.7e+07	1.4e+06/ 1.0e+05	3.7e+02/ 1.4e+02	2.9e+05/ 2.7e+03	3.5e+02/ 1.1e+02	4.7e+06/ 3.5e+04
latencyModelPass	F	5.9e+07/ 1.3e+07	6.2e+07 /6.9e+07	1.2e+06/ 3.7e+04	2.4e+02/ 1.0e+02	2.0e+05/ 4.1e+03	2.2e+02/ 7.2e+01	4.8e+06/ 6.2e+05
memoryOpStatsPass	F	5.8e+07/ 1.9e+07	8.5e+07 /9.1e+07	6.5e+05 /7.3e+05	3.7e+02/ 1.1e+02	2.9e+04/ 2.6e+04	3.4e+02/ 1.0e+02	4.7e+06/ 4.1e+05
stripSideEffectsPass	M	1.5e+08 /2.0e+08	1.8e+08 /6.0e+08	1.9e+06/ 1.3e+06	1.0e+05 /1.4e+05	7.1e+03 /1.7e+04	2.5e+03 /7.6e+03	4.2e+06/ 1.2e+06
targetReturnPass	M	1.5e+08 /1.9e+08	1.7e+08 /6.3e+08	1.6e+06/ 1.2e+06	9.1e+04 /1.3e+05	5.2e+03 /2.4e+04	3.2e+03 /2.2e+04	3.7e+06/ 1.5e+06
throughputModelPass	F	5.9e+07/ 1.5e+07	6.2e+07 /6.9e+07	1.2e+06/ 3.3e+04	3.2e+02/ 1.2e+02	2.0e+05/ 3.6e+03	2.9e+02/ 1.0e+02	4.8e+06/ 6.8e+05
verifyIR	F	3.6e+07/ 1.7e+07	7.4e+07 /8.6e+07	1.1e+06/ 7.9e+05	8.8e+02/ 7.1e+02	2.6e+05/ 2.8e+04	7.9e+02/ 5.4e+02	4.6e+06/ 4.2e+05
Total		8.8e+08/5.8e+08	9.9e+08/1.9e+09	1.3e+07/4.9e+06	2.9e+05/3.3e+05	1.4e+06/1.4e+05	1.1e+04/3.1e+04	5.1e+07/6.9e+06

Table 8: Per-pass PAPI counters for **Compiler** (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
blockCountPass	F	4.1e+08/−	1.6e+08/−	2.3e+06/−	5.0e+02/−	9.2e+04/−	4.9e+02/−	1.4e+07/−
branchStatsPass	F	4.1e+08/−	1.8e+08/−	2.1e+06/−	5.3e+02/−	4.4e+04/−	5.2e+02/−	1.4e+07/−
castInstCountPass	F	4.1e+08/−	1.8e+08/−	2.1e+06/−	4.6e+02/−	4.6e+04/−	4.2e+02/−	1.4e+07/−
has cycle	F	4.7e+08/−	2.1e+08/−	2.5e+06/−	3.3e+02/−	6.9e+04/−	3.2e+02/−	1.7e+07/−
instCountPass	F	4.1e+08/−	1.6e+08/−	2.3e+06/−	3.5e+02/−	9.8e+04/−	3.4e+02/−	1.4e+07/−
latencyModelPass	F	4.3e+08/−	1.7e+08/−	2.1e+06/−	4.7e+02/−	4.9e+04/−	4.5e+02/−	1.4e+07/−
memoryOpStatsPass	F	4.1e+08/−	1.9e+08/−	2.1e+06/−	4.2e+02/−	4.4e+04/−	4.1e+02/−	1.4e+07/−
stripSideEffectsPass	M	6.6e+08/−	3.3e+08/−	8.4e+05/−	1.0e+05/−	1.5e+04/−	1.1e+04/−	2.4e+07/−
targetReturnPass	M	6.5e+08/−	3.2e+08/−	1.1e+06/−	1.2e+05/−	1.7e+04/−	2.2e+04/−	2.2e+07/−
throughputModelPass	F	4.1e+08/−	1.7e+08/−	2.1e+06/−	4.1e+02/−	4.6e+04/−	3.8e+02/−	1.4e+07/−
verifyIR	F	3.5e+08/−	1.2e+08/−	2.5e+06/−	6.9e+02/−	2.1e+05/−	6.0e+02/−	1.1e+07/−
Total		5.0e+09/−	2.2e+09/−	2.2e+07/−	2.2e+05/−	7.3e+05/−	3.7e+04/−	1.7e+08/−

Table 9: Per-pass performance for **DBQuery**, ADT has 15 fields, SoA uses 13 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across $-iterate$ runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
clearQueryFlags	M	14/15	7%	0.0689	0.0784	0.0828	0.2979	0.83 $\times$	1.14 $\times$	0.95 $\times$	0.26 $\times$
countJoins	F	3/15	80%	0.0186	0.0283	0.0115	0.0858	1.62 $\times$	1.52 $\times$	2.45 $\times$	0.33 $\times$
filterSelectivitySkew	F	5/15	67%	0.0188	0.0280	0.0124	0.0867	1.51 $\times$	1.49 $\times$	2.25 $\times$	0.32 $\times$
hashJoinPressure	F	5/15	67%	0.0190	0.0320	0.0126	0.0866	1.50 $\times$	1.69 $\times$	2.54 $\times$	0.37 $\times$
scaleCosts	M	15/15	0%	0.0684	0.0777	0.0843	0.2962	0.81 $\times$	1.14 $\times$	0.92 $\times$	0.26 $\times$
sumCost	F	6/15	60%	0.0309	0.0413	0.0219	0.0901	1.41 $\times$	1.34 $\times$	1.89 $\times$	0.46 $\times$
sumMemory	F	6/15	60%	0.0184	0.0283	0.0151	0.0887	1.22 $\times$	1.54 $\times$	1.88 $\times$	0.32 $\times$
sumRows	F	6/15	60%	0.0267	0.0301	0.0226	0.0873	1.18 $\times$	1.13 $\times$	1.33 $\times$	0.35 $\times$
Total				0.2696	0.3441	0.2633	1.119	1.02 $\times$	1.28 $\times$	1.31 $\times$	0.31 $\times$
Geomean								1.22 $\times$	1.36 $\times$	1.66 $\times$	0.33 $\times$

Table 10: Per-pass PAPI counters for **DBQuery** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
clearQueryFlags	M	1.7e+08 /2.2e+08	1.9e+08 /7.2e+08	2.7e+05 /1.4e+06	8.5e+04 /1.7e+05	3.1e+04/ 2.5e+04	3.1e+03 /9.7e+03	2.4e+06/ 9.1e+05
countJoins	F	9.4e+07/ 5.8e+07	7.4e+07/ 7.1e+07	6.1e+05/ 2.5e+04	3.7e+02/ 2.1e+02	1.4e+05/ 2.0e+03	3.5e+02/ 1.4e+02	2.7e+06/ 5.7e+04
filterSelectivitySkew	F	9.4e+07/ 6.3e+07	1.0e+08 /1.1e+08	5.2e+05/ 8.0e+04	2.8e+02/ 1.5e+02	1.2e+05/ 6.4e+03	2.5e+02/ 1.3e+02	2.7e+06/ 3.2e+05
hashJoinPressure	F	9.6e+07/ 6.4e+07	8.4e+07 /9.9e+07	7.3e+05/ 1.0e+05	3.0e+02/ 1.3e+02	1.9e+05/ 7.4e+03	2.7e+02/ 1.1e+02	2.8e+06/ 1.3e+05
scaleCosts	M	1.6e+08 /2.2e+08	2.2e+08 /7.5e+08	2.5e+05 /1.2e+06	7.4e+04 /1.8e+05	3.4e+04/ 2.6e+04	3.6e+03 /1.6e+04	2.5e+06/ 1.1e+06
sumCost	F	8.5e+07/ 7.5e+07	9.0e+07 /1.2e+08	1.8e+06/ 5.0e+05	5.0e+02/ 3.4e+02	1.9e+05/ 1.7e+04	3.8e+02/ 2.5e+02	2.3e+06/ 3.3e+05
sumMemory	F	9.3e+07/ 7.6e+07	9.0e+07 /1.2e+08	6.7e+05/ 7.9e+04	3.2e+02/ 1.2e+02	1.5e+05/ 1.2e+04	3.1e+02/ 1.0e+02	2.8e+06/ 3.7e+05
sumRows	F	1.3e+08/ 1.1e+08	2.1e+08 /2.3e+08	3.0e+05/ 1.2e+05	2.1e+02 /2.2e+02	2.4e+04/ 7.5e+03	2.0e+02 /2.1e+02	2.4e+06/ 3.1e+05
Total		9.2e+08/8.9e+08	1.1e+09/2.2e+09	5.2e+06/3.5e+06	1.6e+05/3.5e+05	8.8e+05/1.0e+05	8.4e+03/2.7e+04	2.1e+07/3.5e+06

Table 11: Per-pass PAPI counters for **DBQuery** (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
clearQueryFlags	M	2.0e+08 /1.3e+09	3.2e+08 /2.4e+09	6.2e+05 /1.9e+07	9.3e+04 /6.9e+05	1.6e+04 /5.5e+04	3.6e+03 /5.4e+04	2.1e+06/ 1.0e+06
countJoins	F	1.4e+08 /4.4e+08	1.6e+08 /7.7e+08	1.9e+06/ 3.3e+05	3.1e+02 /7.7e+02	1.1e+05/ 1.3e+04	2.9e+02/ 1.8e+02	2.4e+06/ 8.0e+04
filterSelectivitySkew	F	1.4e+08 /4.4e+08	2.0e+08 /7.9e+08	1.4e+06/ 8.7e+05	3.1e+02 /1.2e+03	6.5e+04/ 1.7e+04	2.9e+02 /4.6e+02	2.3e+06/ 2.2e+05
hashJoinPressure	F	1.6e+08 /4.4e+08	1.8e+08 /7.7e+08	1.7e+06/ 6.6e+05	1.8e+02 /8.6e+02	1.1e+05/ 1.6e+04	1.6e+02 /2.6e+02	2.2e+06/ 1.5e+05
scaleCosts	M	2.0e+08 /1.3e+09	3.5e+08 /2.4e+09	5.4e+05 /2.4e+07	7.4e+04 /3.1e+05	1.7e+04 /6.0e+04	3.1e+03 /4.9e+04	2.2e+06/ 1.1e+06
sumCost	F	1.4e+08 /4.5e+08	1.8e+08 /7.8e+08	1.9e+06/ 1.5e+06	4.9e+02 /1.5e+03	8.5e+04/ 2.5e+04	3.9e+02 /7.9e+02	2.1e+06/ 3.2e+05
sumMemory	F	1.4e+08 /4.5e+08	1.8e+08 /7.8e+08	1.5e+06/ 1.2e+06	4.5e+02 /9.4e+02	7.0e+04/ 2.1e+04	4.3e+02 /5.0e+02	2.3e+06/ 3.2e+05
sumRows	F	1.5e+08 /4.4e+08	2.3e+08 /8.3e+08	1.5e+06/ 1.2e+06	3.5e+02 /7.7e+02	5.8e+04/ 2.2e+04	3.4e+02 /4.2e+02	2.3e+06/ 3.5e+05
Total		1.3e+09/5.3e+09	1.8e+09/9.5e+09	1.1e+07/4.9e+07	1.7e+05/1.0e+06	5.3e+05/2.3e+05	8.6e+03/1.1e+05	1.8e+07/3.6e+06

Table 12: Per-pass performance for **DecisionTree**, ADT has 7 fields, SoA uses 6 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across $-iterate$ runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
classify Batch	F	5/7	29%	3.431	4.440	2.582	20.85	1.33 $\times$	1.29 $\times$	1.72 $\times$	0.21 $\times$
classify Depth	F	4/7	43%	0.0475	0.0641	0.0364	0.3042	1.30 $\times$	1.35 $\times$	1.76 $\times$	0.21 $\times$
classify tree	F	5/7	29%	0.0475	0.0635	0.0375	0.2982	1.27 $\times$	1.33 $\times$	1.69 $\times$	0.21 $\times$
countFeatureUses	F	3/7	57%	0.0506	0.1420	0.0323	0.3787	1.57 $\times$	2.81 $\times$	4.40 $\times$	0.38 $\times$
countLeaves	F	2/7	71%	0.0483	0.1405	0.0258	0.3690	1.87 $\times$	2.91 $\times$	5.44 $\times$	0.38 $\times$
countNodes	F	2/7	71%	0.0486	0.1436	0.0272	0.3711	1.79 $\times$	2.95 $\times$	5.28 $\times$	0.39 $\times$
countSmallLeaves	F	3/7	57%	0.0513	0.1367	0.0325	0.3713	1.58 $\times$	2.66 $\times$	4.21 $\times$	0.37 $\times$
inferenceCost	F	2/7	71%	0.0514	0.1391	0.0290	0.3700	1.77 $\times$	2.71 $\times$	4.80 $\times$	0.38 $\times$
sumImpurity	F	3/7	57%	0.0507	0.1395	0.0322	0.4972	1.57 $\times$	2.75 $\times$	4.33 $\times$	0.28 $\times$
sumPathLengths	F	3/7	57%	0.0536	0.1393	0.0339	0.3886	1.58 $\times$	2.60 $\times$	4.11 $\times$	0.36 $\times$
sumSamples	F	3/7	57%	0.0488	0.1353	0.0321	0.3843	1.52 $\times$	2.77 $\times$	4.21 $\times$	0.35 $\times$
treeDepth	F	2/7	71%	0.0516	0.1397	0.0285	0.3697	1.81 $\times$	2.71 $\times$	4.90 $\times$	0.38 $\times$
Total				3.981	5.823	2.929	24.95	1.36 $\times$	1.46 $\times$	1.99 $\times$	0.23 $\times$
Geomean								1.57 $\times$	2.30 $\times$	3.60 $\times$	0.32 $\times$

Table 13: Per-pass PAPI counters for **DecisionTree** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
classify Batch	F	1.7e+10/ 1.3e+10	3.3e+10 /5.9e+10	4.6e+08/ 2.6e+07	1.2e+05/ 1.8e+04	1.6e+06 /2.0e+06	9.6e+04/ 1.8e+04	9.2e+08/ 1.8e+07
classify Depth	F	2.4e+08/ 1.8e+08	4.7e+08 /8.5e+08	6.6e+06/ 3.7e+05	1.1e+03/ 3.4e+02	2.3e+04 /2.8e+04	1.0e+03/ 3.1e+02	1.3e+07/ 3.3e+05
classify tree	F	2.4e+08/ 1.9e+08	4.7e+08 /8.5e+08	6.6e+06/ 3.7e+05	1.1e+03/ 3.0e+02	2.0e+04 /2.8e+04	1.1e+03/ 2.8e+02	1.3e+07/ 3.3e+05
countFeatureUses	F	2.6e+08/ 1.6e+08	7.5e+08 /7.8e+08	7.5e+06/ 2.5e+05	7.2e+02/ 5.8e+02	3.6e+04/ 1.4e+04	7.0e+02/ 5.5e+02	1.5e+07/ 2.1e+06
countLeaves	F	2.5e+08/ 1.3e+08	6.0e+08 /6.1e+08	7.4e+06/ 7.6e+05	1.1e+03/ 7.6e+02	4.6e+04/ 1.7e+04	1.0e+03/ 7.1e+02	1.4e+07/ 2.3e+05
countNodes	F	2.5e+08/ 1.4e+08	6.1e+08 /6.2e+08	7.1e+06/ 7.6e+05	1.5e+03/ 8.1e+02	5.2e+04/ 1.7e+04	1.3e+03/ 6.8e+02	1.4e+07/ 2.3e+05
countSmallLeaves	F	2.6e+08/ 1.6e+08	7.1e+08 /8.4e+08	5.9e+06/ 9.1e+04	6.3e+02 /7.5e+02	3.4e+04/ 1.3e+04	6.1e+02 /7.2e+02	1.5e+07/ 2.0e+06
inferenceCost	F	2.6e+08/ 1.5e+08	6.1e+08/ 6.1e+08	8.2e+06/ 1.2e+05	1.2e+03/ 6.8e+02	1.7e+04/ 8.0e+03	1.2e+03/ 6.5e+02	1.4e+07/ 4.2e+05
sumImpurity	F	2.6e+08/ 1.6e+08	7.6e+08 /7.8e+08	7.6e+06/ 5.1e+05	1.2e+03/ 6.4e+02	3.6e+04/ 1.3e+04	1.2e+03/ 6.0e+02	1.4e+07/ 2.1e+06
sumPathLengths	F	2.7e+08/ 1.7e+08	8.0e+08 /9.0e+08	7.5e+06/ 3.3e+05	1.4e+03/ 6.6e+02	4.0e+04/ 1.4e+04	1.4e+03/ 6.4e+02	1.5e+07/ 2.0e+06
sumSamples	F	2.5e+08/ 1.6e+08	6.4e+08 /7.7e+08	6.3e+06/ 2.3e+05	7.7e+02 /8.1e+02	4.2e+04/ 1.4e+04	7.5e+02 /7.9e+02	1.5e+07/ 2.1e+06
treeDepth	F	2.6e+08/ 1.4e+08	6.1e+08/ 6.1e+08	8.1e+06/ 6.7e+05	1.2e+03/ 7.5e+02	1.9e+04/ 1.6e+04	1.1e+03/ 7.0e+02	1.4e+07/ 2.6e+05
Total		2.0e+10/1.5e+10	4.0e+10/6.8e+10	5.4e+08/3.1e+07	1.3e+05/2.5e+04	1.9e+06/2.1e+06	1.1e+05/2.4e+04	1.1e+09/3.0e+07

Table 14: Per-pass PAPI counters for **DecisionTree** (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
classify Batch	F	2.2e+10 /1.1e+11	5.1e+10 /1.9e+11	1.5e+07 /4.7e+07	2.1e+04 /1.4e+05	3.6e+05 /2.2e+06	2.1e+04 /2.2e+04	8.4e+08/ 1.9e+07
classify Depth	F	3.2e+08 /1.5e+09	7.4e+08 /2.7e+09	2.5e+05 /6.5e+05	4.4e+02 /1.8e+03	5.2e+03 /3.1e+04	4.3e+02 /4.9e+02	1.2e+07/ 3.4e+05
classify tree	F	3.2e+08 /1.5e+09	7.4e+08 /2.7e+09	2.5e+05 /6.8e+05	4.2e+02 /2.4e+03	5.3e+03 /3.2e+04	4.0e+02 /8.0e+02	1.2e+07/ 3.3e+05
countFeatureUses	F	7.2e+08 /1.9e+09	1.3e+09 /3.1e+09	2.0e+06 /3.2e+06	3.1e+02 /2.7e+03	5.2e+03 /7.0e+04	2.9e+02 /1.7e+03	9.3e+06/ 1.2e+06
countLeaves	F	7.1e+08 /1.9e+09	1.2e+09 /2.9e+09	4.4e+06/ 6.7e+05	6.5e+02 /2.7e+03	6.2e+03 /3.2e+04	6.3e+02/ 3.9e+02	9.1e+06/ 3.8e+05
countNodes	F	7.2e+08 /1.9e+09	1.2e+09 /3.0e+09	4.7e+06/ 6.7e+05	9.0e+02 /3.5e+03	1.4e+04 /3.2e+04	8.0e+02/ 6.0e+02	9.1e+06/ 3.8e+05
countSmallLeaves	F	6.9e+08 /1.9e+09	1.3e+09 /2.9e+09	1.1e+06 /3.4e+06	4.8e+02 /2.1e+03	4.2e+03 /7.3e+04	4.5e+02 /8.1e+02	9.2e+06/ 1.2e+06
inferenceCost	F	7.1e+08 /1.9e+09	1.3e+09 /3.0e+09	4.4e+06/ 6.1e+05	5.3e+02 /2.0e+03	4.8e+03 /3.0e+04	5.0e+02 /7.0e+02	9.1e+06/ 3.8e+05
sumImpurity	F	7.1e+08 /1.9e+09	1.3e+09 /3.0e+09	2.2e+06 /3.6e+06	5.4e+02 /1.1e+03	6.5e+03 /7.6e+04	5.2e+02 /8.6e+02	9.4e+06/ 1.2e+06
sumPathLengths	F	7.1e+08 /2.0e+09	1.3e+09 /3.1e+09	1.5e+06 /3.6e+06	5.0e+02 /4.6e+03	4.0e+03 /7.9e+04	4.8e+02 /2.0e+03	9.1e+06/ 1.2e+06
sumSamples	F	6.9e+08 /2.0e+09	1.2e+09 /3.0e+09	1.1e+06 /3.6e+06	8.4e+02 /2.4e+03	4.4e+03 /7.7e+04	8.2e+02 /1.2e+03	9.2e+06/ 1.2e+06
treeDepth	F	7.1e+08 /1.9e+09	1.3e+09 /3.0e+09	4.4e+06/ 6.2e+05	4.8e+02 /2.6e+03	5.4e+03 /3.1e+04	4.4e+02 /5.5e+02	9.1e+06/ 4.0e+05
Total		2.9e+10/1.3e+11	6.4e+10/2.2e+11	4.1e+07/6.9e+07	2.7e+04/1.6e+05	4.2e+05/2.8e+06	2.7e+04/3.2e+04	9.4e+08/2.7e+07

Table 15: Per-pass performance for **DomTree**, ADT has 15 fields, SoA uses 14 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
SumArea	F	6/15	60%	0.0355	0.0696	0.0354	0.0909	1.00 $\times$	1.96$\times$	1.96$\times$	0.77 $\times$
computeWidths	M	14/15	7%	0.0299	0.0307	0.0444	0.0877	0.67 $\times$	1.02 $\times$	0.69 $\times$	0.35 $\times$
count styled	F	3/15	80%	0.0336	0.0680	0.0116	0.0861	2.90$\times$	2.02$\times$	5.86$\times$	0.79 $\times$
find max Bottom	F	5/15	67%	0.0337	0.0678	0.0194	0.0871	1.73$\times$	2.01$\times$	3.49$\times$	0.78 $\times$
scaleLayout	M	15/15	0%	0.0289	0.0304	0.0365	0.2142	0.79 $\times$	1.05 $\times$	0.83 $\times$	0.14 $\times$
sumTextWidth	F	3/15	80%	0.0333	0.0686	0.0115	0.0861	2.89$\times$	2.06$\times$	5.95$\times$	0.80 $\times$
Total				0.1950	0.3350	0.1589	0.6521	1.23$\times$	1.72$\times$	2.11$\times$	0.51 $\times$
Geomean								1.41$\times$	1.61$\times$	2.27$\times$	0.51 $\times$

Table 16: Per-pass PAPI counters for **DomTree** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
SumArea	F	1.8e+08/ 9.5e+07	2.9e+08 /4.9e+08	2.3e+06/ 9.6e+05	9.6e+02/ 4.4e+02	1.7e+05/ 2.0e+04	7.4e+02/ 3.6e+02	1.4e+07/ 1.3e+06
computeWidths	M	6.6e+07 /1.4e+08	1.4e+08 /4.6e+08	1.2e+06 /1.5e+06	5.2e+04 /7.2e+04	4.6e+03 /1.0e+04	1.6e+03 /1.7e+04	1.3e+06/ 4.6e+05
count styled	F	1.7e+08/ 5.9e+07	2.6e+08 /2.6e+08	2.1e+06/ 2.4e+04	6.6e+02/ 1.5e+02	3.1e+05/ 4.9e+03	6.3e+02/ 1.3e+02	1.4e+07/ 7.2e+05
find max Bottom	F	1.7e+08/ 9.1e+07	2.5e+08 /4.1e+08	2.3e+06/ 8.7e+05	4.3e+02/ 1.4e+02	1.6e+05/ 9.2e+03	3.8e+02/ 1.1e+02	1.4e+07/ 1.4e+06
scaleLayout	M	5.9e+07 /9.9e+07	8.8e+07 /2.9e+08	1.1e+06 /2.3e+06	4.6e+04 /5.0e+04	2.4e+04/ 9.3e+03	2.5e+03 /1.5e+04	1.4e+06/ 4.4e+05
sumTextWidth	F	1.7e+08/ 5.9e+07	2.0e+08 /2.4e+08	2.4e+06/ 2.0e+04	8.3e+02/ 1.4e+02	4.5e+05/ 4.2e+03	8.0e+02/ 1.2e+02	1.4e+07/ 7.2e+05
Total		8.2e+08/5.4e+08	1.2e+09/2.2e+09	1.1e+07/5.7e+06	1.0e+05/1.2e+05	1.1e+06/5.8e+04	6.6e+03/3.2e+04	5.7e+07/5.0e+06

Table 17: Per-pass PAPI counters for **DomTree** (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
SumArea	F	3.5e+08 /4.5e+08	4.4e+08 /1.8e+09	8.1e+06/ 2.5e+06	6.2e+02 /1.4e+03	1.8e+04 /4.1e+04	4.9e+02 /1.3e+03	1.0e+07/ 1.4e+06
computeWidths	M	6.8e+07 /3.6e+08	1.5e+08 /9.0e+08	1.3e+06 /3.6e+06	4.6e+04 /2.4e+05	3.8e+03 /1.4e+04	1.1e+03 /2.3e+04	1.2e+06/ 5.1e+05
count styled	F	3.5e+08 /4.4e+08	4.3e+08 /1.7e+09	9.9e+06/ 1.2e+06	2.0e+02 /5.0e+02	4.5e+04/ 3.6e+04	1.8e+02 /4.5e+02	1.1e+07/ 4.6e+05
find max Bottom	F	3.4e+08 /4.4e+08	4.6e+08 /1.8e+09	1.0e+07/ 1.4e+06	1.2e+02 /7.5e+02	2.0e+04 /2.7e+04	1.1e+02 /7.4e+02	1.0e+07/ 1.0e+06
scaleLayout	M	6.8e+07 /3.6e+08	1.1e+08 /7.6e+08	9.6e+05 /4.1e+06	5.4e+04 /9.3e+04	8.8e+03 /1.2e+04	1.3e+03 /2.6e+04	1.3e+06/ 4.9e+05
sumTextWidth	F	3.5e+08 /4.4e+08	3.9e+08 /1.7e+09	1.0e+07/ 1.3e+06	2.8e+02 /1.2e+03	9.3e+04/ 4.0e+04	2.6e+02 /9.3e+02	1.0e+07/ 4.4e+05
Total		1.5e+09/2.5e+09	2.0e+09/8.6e+09	4.1e+07/1.4e+07	1.0e+05/3.4e+05	1.9e+05/1.7e+05	3.4e+03/5.3e+04	4.4e+07/4.3e+06

Table 18: Per-pass performance for **KDTree**, ADT has 17 fields, SoA uses 16 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
Find nearest Neighbour	F	13/17	24%	0.0277	0.0465	0.0249	0.1269	1.11$\times$	1.68$\times$	1.87$\times$	0.37 $\times$
countInRange tight_box	F	13/17	24%	0.0272	0.0396	0.0198	0.1325	1.37$\times$	1.46$\times$	2.00$\times$	0.30 $\times$
photonMappingBenchmark	F	12/17	29%	0.0418	0.0473	0.0409	0.1313	1.02 $\times$	1.13$\times$	1.16$\times$	0.36 $\times$
pointCloudNeighborhood	F	11/17	35%	0.0249	0.0381	0.0165	0.1230	1.51$\times$	1.53$\times$	2.31$\times$	0.31 $\times$
sumMassInRange	F	14/17	18%	0.0265	0.0397	0.0203	0.1332	1.31$\times$	1.50$\times$	1.96$\times$	0.30 $\times$
twoPointCorrelation bin_8_16	F	11/17	35%	0.0281	0.0391	0.0207	0.1229	1.35$\times$	1.39$\times$	1.89$\times$	0.32 $\times$
Total				0.1762	0.2505	0.1432	0.7698	1.23$\times$	1.42$\times$	1.75$\times$	0.33 $\times$
Geomean								1.27$\times$	1.44$\times$	1.82$\times$	0.32 $\times$

Table 19: Per-pass PAPI counters for **KDTree** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
Find nearest Neighbour	F	1.4e+08/ 1.1e+08	4.7e+08 /6.3e+08	2.7e+06/ 2.0e+06	7.6e+02/ 4.9e+02	4.7e+03 /2.4e+04	6.2e+02/ 3.8e+02	7.7e+06/ 1.6e+06
countInRange tight_box	F	1.4e+08/ 1.0e+08	4.0e+08 /5.5e+08	2.2e+06/ 2.1e+06	2.1e+02/ 1.5e+02	3.4e+03 /1.8e+04	1.8e+02/ 1.4e+02	7.7e+06/ 1.8e+06
photonMappingBenchmark	F	2.1e+08/ 2.1e+08	1.0e+09 /1.2e+09	2.1e+06 /2.4e+06	4.4e+02 /6.3e+02	3.5e+03 /1.5e+04	3.7e+02 /6.1e+02	5.6e+06/ 1.5e+06
pointCloudNeighborhood	F	1.3e+08/ 8.4e+07	3.3e+08 /4.8e+08	2.2e+06/ 9.1e+05	1.8e+02/1.8e+02	7.3e+03 /1.6e+04	1.6e+02 /1.7e+02	7.8e+06/ 1.6e+06
sumMassInRange	F	1.3e+08/ 1.0e+08	4.0e+08 /5.7e+08	2.5e+06 /2.7e+06	2.6e+02/ 2.1e+02	4.1e+03 /2.0e+04	2.0e+02/ 2.0e+02	7.7e+06/ 2.0e+06
twoPointCorrelation bin_8_16	F	1.4e+08/ 1.0e+08	4.8e+08 /6.1e+08	2.1e+06/ 7.0e+05	2.3e+02/ 1.8e+02	3.2e+03 /1.3e+04	2.2e+02/ 1.6e+02	7.7e+06/ 1.4e+06
Total		8.9e+08/7.1e+08	3.1e+09/4.0e+09	1.4e+07/1.1e+07	2.1e+03/1.8e+03	2.6e+04/1.1e+05	1.7e+03/1.7e+03	4.4e+07/9.8e+06

Table 20: Per-pass PAPI counters for **KDTree** (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
Find nearest Neighbour	F	2.0e+08 /6.4e+08	4.9e+08 /1.3e+09	5.9e+06/ 5.4e+06	5.5e+02 /2.5e+03	1.2e+05/ 2.6e+04	3.7e+02 /2.3e+03	5.4e+06/ 1.5e+06
countInRange tight_box	F	2.0e+08 /6.7e+08	4.5e+08 /1.3e+09	5.1e+06 /5.9e+06	1.8e+02 /2.7e+03	4.9e+03 /2.2e+04	1.5e+02 /2.7e+03	5.7e+06/ 1.6e+06
photonMappingBenchmark	F	2.4e+08 /6.7e+08	1.1e+09 /2.0e+09	2.9e+06 /6.0e+06	5.4e+02 /1.8e+03	3.2e+03 /2.2e+04	5.2e+02 /1.8e+03	5.2e+06/ 1.5e+06
pointCloudNeighborhood	F	1.9e+08 /6.2e+08	4.1e+08 /1.3e+09	5.1e+06/ 4.8e+06	2.4e+02 /1.5e+03	5.7e+03 /1.9e+04	2.2e+02 /1.5e+03	5.9e+06/ 1.4e+06
sumMassInRange	F	2.0e+08 /6.8e+08	4.7e+08 /1.3e+09	5.1e+06 /6.4e+06	3.6e+02 /2.0e+03	4.4e+03 /2.5e+04	3.3e+02 /1.9e+03	5.7e+06/ 1.6e+06
twoPointCorrelation bin_8_16	F	2.0e+08 /6.2e+08	6.1e+08 /1.4e+09	4.9e+06/ 3.6e+06	2.9e+02 /1.9e+03	3.3e+04/ 1.7e+04	2.6e+02 /1.8e+03	5.8e+06/ 1.4e+06
Total		1.2e+09/3.9e+09	3.5e+09/8.4e+09	2.9e+07/3.2e+07	2.2e+03/1.2e+04	1.7e+05/1.3e+05	1.9e+03/1.2e+04	3.4e+07/8.9e+06

Table 21: Per-pass performance for **LinearListReduction**, ADT has 11 fields, SoA uses 11 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
reduction	F	2/11	82%	0.0303	0.1396	0.0065	<i>OOM</i>	4.63 $\times$	4.60 $\times$	21.33 $\times$	–
Total				0.0303	0.1396	0.0065	–	4.63 $\times$	4.60 $\times$	21.33 $\times$	–
Geomean								4.63 $\times$	4.60 $\times$	21.33 $\times$	–

Table 22: Per-pass PAPI counters for **LinearListReduction** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
reduction	F	1.4e+08/ 2.5e+07	9.0e+07 /1.0e+08	1.5e+06/ 1.2e+06	7.4e+02/ 3.0e+02	3.0e+05/ 6.8e+04	6.4e+02/ 1.8e+02	1.2e+07/ 1.1e+06
Total		1.4e+08/2.5e+07	9.0e+07/1.0e+08	1.5e+06/1.2e+06	7.4e+02/3.0e+02	3.0e+05/6.8e+04	6.4e+02/1.8e+02	1.2e+07/1.1e+06

Table 23: Per-pass PAPI counters for **LinearListReduction** (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
reduction	F	7.0e+08/–	2.5e+08/–	5.0e+06/–	1.0e+03/–	5.5e+05/–	8.1e+02/–	2.6e+07/–
Total		7.0e+08/–	2.5e+08/–	5.0e+06/–	1.0e+03/–	5.5e+05/–	8.1e+02/–	2.6e+07/–

Table 24: Per-pass performance for **List**, ADT has 2 fields, SoA uses 2 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
add1 List	M	2/2	0%	0.2556	<i>OOM</i>	0.3424	<i>OOM</i>	0.75 $\times$	–	–	–
length List	F	1/2	50%	0.0420	<i>OOM</i>	0.0216	<i>OOM</i>	1.94 $\times$	–	–	–
sumList	F	2/2	0%	0.0440	<i>OOM</i>	0.0456	<i>OOM</i>	0.96 $\times$	–	–	–
sumList tail recursive	F	2/2	0%	0.0442	<i>OOM</i>	0.0446	<i>OOM</i>	0.99 $\times$	–	–	–
Total				0.3857	–	0.4542	–	0.85 $\times$	–	–	–
Geomean								1.08 $\times$	–	–	–

Table 25: Per-pass PAPI counters for **List** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
add1 List	M	5.9e+08 /9.9e+08	1.9e+09 /2.8e+09	7.9e+06/ 4.0e+06	2.6e+05 /5.5e+05	1.1e+04 /9.1e+04	6.6e+03 /1.2e+04	9.7e+06/ 7.6e+06
length List	F	2.1e+08/ 1.1e+08	8.0e+08/ 8.0e+08	1.2e+05/ 1.1e+05	1.5e+02/ 1.0e+02	2.9e+03 /1.4e+04	1.3e+02/ 8.5e+01	1.3e+07/ 1.2e+06
sumList	F	2.2e+08 /2.3e+08	9.0e+08 /1.0e+09	7.4e+05/ 5.4e+05	1.3e+02 /3.3e+02	3.4e+03 /6.8e+04	1.1e+02 /3.2e+02	1.3e+07/ 1.2e+07
sumList tail recursive	F	2.2e+08 /2.3e+08	9.0e+08 /1.0e+09	5.7e+05/ 5.7e+05	1.8e+02 /2.1e+02	3.1e+03 /5.9e+04	1.6e+02 /1.9e+02	1.3e+07/ 1.2e+07
Total		1.3e+09/1.6e+09	4.5e+09/5.6e+09	9.4e+06/5.2e+06	2.6e+05/5.5e+05	2.1e+04/2.3e+05	7.0e+03/1.3e+04	4.9e+07/3.3e+07

Table 26: Per-pass PAPI counters for **List** (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
Total		–/–	–/–	–/–	–/–	–/–	–/–	–/–

Table 27: Per-pass performance for **MonoTree**, ADT has 3 fields, SoA uses 2 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
add1Tree	M	3/3	0%	0.0945	0.1045	0.0752	0.1215	1.26 $\times$	1.11 $\times$	1.39 $\times$	0.86 $\times$
sumTree	F	3/3	0%	0.0091	0.0443	0.0113	0.0815	0.80 $\times$	4.89 $\times$	3.93 $\times$	0.54 $\times$
sumTree TailRec	F	3/3	0%	0.0084	0.0445	0.0110	0.0814	0.76 $\times$	5.30 $\times$	4.03 $\times$	0.55 $\times$
Total				0.1120	0.1934	0.0975	0.2844	1.15 $\times$	1.73 $\times$	1.98 $\times$	0.68 $\times$
Geomean								0.92 $\times$	3.06 $\times$	2.80 $\times$	0.63 $\times$

Table 28: Per-pass PAPI counters for `MonoTree` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
add1Tree	M	1.7e+08/ 1.5e+08	5.0e+08 /5.9e+08	7.2e+05/ 5.1e+05	3.3e+04 /4.7e+04	1.2e+04/ 6.9e+03	1.8e+03 /2.6e+03	4.6e+05/ 4.3e+05
sumTree	F	4.6e+07 /5.7e+07	2.2e+08 /2.6e+08	6.4e+03 /3.4e+04	2.8e+02/ 2.5e+02	6.3e+02 /3.8e+03	2.5e+02/ 2.3e+02	8.8e+05/ 6.7e+05
sumTree TailRec	F	4.3e+07 /5.6e+07	1.9e+08 /2.2e+08	5.9e+03 /1.7e+04	2.7e+02 /3.2e+02	7.1e+02 /3.8e+03	2.4e+02 /3.0e+02	8.8e+05/ 6.7e+05
Total		2.5e+08/2.6e+08	9.1e+08/1.1e+09	7.4e+05/5.6e+05	3.3e+04/4.8e+04	1.3e+04/1.5e+04	2.3e+03/3.1e+03	2.2e+06/1.8e+06

Table 29: Per-pass PAPI counters for `MonoTree` (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
add1Tree	M	2.5e+08 /5.1e+08	7.1e+08 /1.2e+09	1.4e+06 /1.4e+06	3.4e+04 /5.3e+04	2.2e+04 /2.5e+04	2.1e+03 /2.9e+03	3.6e+05/ 3.5e+05
sumTree	F	2.2e+08 /4.1e+08	4.3e+08 /6.3e+08	1.3e+06 /1.3e+06	2.7e+02 /2.7e+02	2.1e+04 /2.4e+04	2.3e+02 /2.4e+02	3.3e+05 /3.4e+05
sumTree TailRec	F	2.3e+08 /4.1e+08	4.1e+08 /6.4e+08	1.3e+06 /1.3e+06	3.2e+02 /3.4e+02	2.1e+04 /2.4e+04	2.8e+02 /3.3e+02	3.3e+05 /3.4e+05
Total		7.0e+08/1.3e+09	1.6e+09/2.5e+09	4.0e+06/4.1e+06	3.5e+04/5.3e+04	6.5e+04/7.3e+04	2.6e+03/3.5e+03	1.0e+06/1.0e+06

Table 30: Per-pass performance for `ObjectGraph`, ADT has 5 fields, SoA uses 4 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across -iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
countLargeItems	F	3/5	40%	0.0124	0.0441	0.0111	0.0466	1.12 $\times$	3.56 $\times$	3.99 $\times$	0.95 $\times$
countMarked	F	3/5	40%	0.0124	0.0442	0.0139	0.0476	0.89 $\times$	3.56 $\times$	3.17 $\times$	0.93 $\times$
countSurvivors	F	4/5	20%	0.0134	0.0448	0.0130	0.0468	1.03 $\times$	3.35 $\times$	3.44 $\times$	0.96 $\times$
deadBytes	F	4/5	20%	0.0131	0.0439	0.0124	0.0471	1.05 $\times$	3.36 $\times$	3.54 $\times$	0.93 $\times$
liveBytes	F	4/5	20%	0.0130	0.0430	0.0125	0.0471	1.04 $\times$	3.32 $\times$	3.46 $\times$	0.91 $\times$
sumObjIds	F	3/5	40%	0.0124	0.0429	0.0106	0.0472	1.17 $\times$	3.45 $\times$	4.04 $\times$	0.91 $\times$
sweepUnmarked	M	5/5	0%	0.0722	0.0879	0.0808	0.1847	0.89 $\times$	1.22 $\times$	1.09 $\times$	0.48 $\times$
totalHeapSize	F	3/5	40%	0.0123	0.0639	0.0192	0.0485	0.64 $\times$	5.18 $\times$	3.33 $\times$	1.32 $\times$
touchHotObjects	M	5/5	0%	0.0706	0.0871	0.0808	0.1890	0.87 $\times$	1.23 $\times$	1.08 $\times$	0.46 $\times$
Total				0.2317	0.5018	0.2542	0.7045	0.91 $\times$	2.17 $\times$	1.97 $\times$	0.71 $\times$
Geomean								0.96 $\times$	2.86 $\times$	2.73 $\times$	0.83 $\times$

Table 31: Per-pass PAPI counters for `ObjectGraph` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
countLargeItems	F	6.3e+07/ 5.4e+07	2.6e+08 /2.8e+08	1.3e+06/ 6.5e+04	4.8e+02/ 4.1e+02	3.3e+03 /3.8e+03	4.5e+02/ 3.6e+02	3.0e+06/ 7.0e+05
countMarked	F	6.3e+07/ 5.3e+07	2.6e+08 /2.8e+08	1.3e+06/ 5.0e+04	3.8e+02/ 3.2e+02	3.2e+03 /3.9e+03	3.4e+02/ 2.9e+02	3.0e+06/ 6.8e+05
countSurvivors	F	6.8e+07/ 6.6e+07	3.1e+08 /3.8e+08	9.0e+05/ 6.9e+04	3.6e+02/ 2.8e+02	2.3e+03 /5.6e+03	3.3e+02/ 2.5e+02	2.9e+06/ 1.2e+06
deadBytes	F	6.6e+07/ 6.3e+07	2.7e+08 /3.5e+08	7.7e+05/ 5.5e+04	2.4e+02 /3.2e+02	2.3e+03 /5.3e+03	2.3e+02 /2.9e+02	2.9e+06/ 1.3e+06
liveBytes	F	6.6e+07/ 6.3e+07	2.7e+08 /3.5e+08	9.0e+05/ 5.3e+04	2.3e+02 /2.7e+02	2.3e+03 /5.5e+03	2.1e+02 /2.4e+02	2.9e+06/ 1.3e+06
sumObjIds	F	6.3e+07/ 5.4e+07	2.5e+08 /2.7e+08	1.2e+06/ 6.8e+04	2.9e+02/ 2.6e+02	2.6e+03 /4.6e+03	2.8e+02/ 2.4e+02	2.9e+06/ 6.9e+05
sweepUnmarked	M	1.9e+08 /2.4e+08	5.5e+08 /8.8e+08	1.1e+06/ 4.7e+05	9.5e+04 /1.1e+05	7.1e+03 /1.1e+04	4.5e+03 /5.9e+03	2.0e+06/ 9.9e+05
totalHeapSize	F	6.2e+07/ 5.3e+07	2.5e+08 /2.7e+08	1.3e+06/ 7.9e+05	8.1e+02 /8.8e+02	2.6e+03 /1.6e+04	6.8e+02 /7.4e+02	3.0e+06/ 5.5e+05
touchHotObjects	M	1.9e+08 /2.3e+08	5.7e+08 /9.0e+08	1.2e+06/ 4.8e+05	1.0e+05 /1.3e+05	4.3e+03 /1.1e+04	4.1e+03 /5.9e+03	1.9e+06/ 1.0e+06
Total		8.3e+08/8.7e+08	3.0e+09/4.0e+09	1.0e+07/2.1e+06	2.0e+05/2.4e+05	3.0e+04/6.7e+04	1.1e+04/1.4e+04	2.4e+07/8.4e+06

Table 32: Per-pass PAPI counters for `ObjectGraph` (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
countLargeItems	F	2.2e+08 /2.4e+08	4.6e+08 /7.9e+08	5.5e+05/ 4.8e+05	3.5e+02/ 1.8e+02	4.8e+03 /8.2e+03	3.1e+02/ 1.8e+02	1.8e+06/ 4.2e+05
countMarked	F	2.2e+08 /2.4e+08	4.6e+08 /7.9e+08	5.4e+05 /1.0e+06	2.0e+02 /2.5e+02	4.8e+03 /2.0e+04	1.7e+02 /2.1e+02	1.8e+06/ 3.9e+05
countSurvivors	F	2.3e+08 /2.4e+08	5.5e+08 /8.2e+08	1.8e+05 /4.0e+05	2.3e+02 /3.1e+02	1.9e+03 /6.5e+03	2.1e+02 /2.9e+02	1.8e+06/ 7.3e+05
deadBytes	F	2.2e+08 /2.4e+08	4.9e+08 /8.1e+08	3.6e+05 /4.4e+05	1.5e+02 /4.2e+02	3.6e+03 /6.3e+03	1.0e+02 /3.9e+02	1.8e+06/ 7.3e+05
liveBytes	F	2.2e+08 /2.4e+08	5.0e+08 /8.1e+08	1.5e+05 /4.1e+05	1.6e+02 /3.8e+02	1.9e+03 /6.4e+03	1.4e+02 /3.7e+02	1.8e+06/ 7.3e+05
sumObjIds	F	2.2e+08 /2.4e+08	4.4e+08 /7.8e+08	5.4e+05/ 5.0e+05	3.4e+02/ 2.3e+02	4.5e+03 /9.3e+03	3.0e+02/ 2.2e+02	1.8e+06/ 4.1e+05
sweepUnmarked	M	2.7e+08 /7.6e+08	7.6e+08 /2.1e+09	1.8e+05 /2.4e+06	7.9e+04 /1.3e+05	3.4e+03 /1.6e+04	2.3e+03 /4.9e+03	1.8e+06/ 9.2e+05
totalHeapSize	F	2.2e+08 /2.5e+08	4.4e+08 /7.8e+08	6.5e+05 /1.3e+06	5.8e+02 /8.6e+02	6.7e+03 /2.6e+04	4.2e+02 /6.8e+02	1.7e+06/ 3.5e+05
touchHotObjects	M	2.7e+08 /7.8e+08	7.7e+08 /2.2e+09	2.4e+05 /1.7e+06	8.6e+04 /1.1e+05	2.9e+03 /1.8e+04	2.4e+03 /5.9e+03	1.7e+06/ 9.4e+05
Total		2.1e+09/3.2e+09	4.9e+09/9.9e+09	3.4e+06/8.6e+06	1.7e+05/2.4e+05	3.5e+04/1.2e+05	6.4e+03/1.3e+04	1.6e+07/5.6e+06

Table 33: Per-pass performance for `PiecewiseFunctions`, ADT has 8 fields, SoA uses 7 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across -iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
addConstPW	M	8/8	0%	0.1185	0.1334	0.1328	0.4058	0.89 $\times$	1.13 $\times$	1.00 $\times$	0.33 $\times$
autorefineMaxLevel	F	4/8	50%	0.0200	0.0455	0.0189	0.2369	1.06 $\times$	2.28 $\times$	2.41 $\times$	0.19 $\times$
compressMass	F	3/8	62%	0.0191	0.0460	0.0114	0.2345	1.68 $\times$	2.41 $\times$	4.04 $\times$	0.20 $\times$
diffPW	M	8/8	0%	0.1200	0.1320	0.1287	0.4056	0.93 $\times$	1.10 $\times$	1.03 $\times$	0.33 $\times$
lbDeuxLoadProxy	F	5/8	38%	0.0209	0.0479	0.0186	0.2372	1.12 $\times$	2.29 $\times$	2.57 $\times$	0.20 $\times$
norm2Estimate	F	5/8	38%	0.0247	0.0622	0.0229	0.2012	1.08 $\times$	2.52 $\times$	2.72 $\times$	0.31 $\times$
pmapCutHistogram	F	4/8	50%	0.0317	0.0561	0.0126	0.2431	2.52 $\times$	1.77 $\times$	4.46 $\times$	0.23 $\times$
truncateTolViolations	F	3/8	62%	0.0193	0.0468	0.0115	0.2050	1.68 $\times$	2.43 $\times$	4.07 $\times$	0.23 $\times$
Total				0.3741	0.5699	0.3574	2.169	1.05 $\times$	1.52 $\times$	1.59 $\times$	0.26 $\times$
Geomean								1.29 $\times$	1.90 $\times$	2.44 $\times$	0.25 $\times$

Table 34: Per-pass PAPI counters for `PiecewiseFunctions` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
addConstPW	M	2.7e+08 /3.3e+08	5.7e+08 /1.2e+09	1.5e+06 /1.7e+06	1.3e+05 /2.6e+05	1.4e+04 /2.6e+04	7.8e+03 /2.2e+04	4.6e+06/ 1.8e+06
autorefineMaxLevel	F	1.0e+08/ 9.6e+07	2.5e+08 /3.7e+08	2.5e+06/ 2.5e+05	7.2e+02/ 1.3e+02	6.9e+03/ 6.4e+03	7.1e+02/ 1.2e+02	6.2e+06/ 1.0e+06
compressMass	F	9.7e+07/ 5.8e+07	2.5e+08 /2.8e+08	2.1e+06/ 2.8e+04	7.3e+02/ 3.4e+02	1.9e+04/ 5.1e+03	7.2e+02/ 3.2e+02	6.4e+06/ 7.0e+05
diffPW	M	2.8e+08 /3.2e+08	5.9e+08 /1.3e+09	8.2e+05 /1.8e+06	1.5e+05 /1.9e+05	1.0e+04 /2.3e+04	8.4e+03 /1.9e+04	4.2e+06/ 1.8e+06
lbDeuxLoadProxy	F	1.0e+08/ 9.5e+07	3.3e+08 /4.9e+08	2.4e+06/ 3.3e+05	6.4e+02/ 2.2e+02	1.5e+04/ 9.1e+03	6.2e+02/ 2.0e+02	6.3e+06/ 1.6e+06
norm2Estimate	F	1.2e+08/ 9.0e+07	3.4e+08 /4.4e+08	1.5e+06/ 7.7e+05	8.7e+02/ 6.6e+02	1.1e+04 /1.5e+04	7.5e+02/ 5.4e+02	6.2e+06/ 1.3e+06
pmapCutHistogram	F	1.6e+08/ 6.4e+07	2.8e+08 /3.7e+08	1.7e+06/ 5.6e+04	1.5e+03/ 4.2e+02	1.0e+05/ 7.4e+03	1.5e+03/ 4.0e+02	4.8e+06/ 1.3e+06
truncateTolViolations	F	9.8e+07/ 5.8e+07	2.5e+08 /2.9e+08	2.1e+06/ 2.7e+04	8.6e+02/ 3.6e+02	1.7e+04/ 4.9e+03	8.4e+02/ 3.4e+02	6.4e+06/ 7.1e+05
Total		1.2e+09/1.1e+09	2.8e+09/4.7e+09	1.4e+07/5.0e+06	2.8e+05/4.5e+05	2.0e+05/9.7e+04	2.1e+04/4.3e+04	4.5e+07/1.0e+07

Table 35: Per-pass PAPI counters for `PiecewiseFunctions` (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
addConstPW	M	3.3e+08 /1.7e+09	7.8e+08 /3.3e+09	3.7e+05 /5.2e+06	1.2e+05 /3.2e+05	6.4e+03 /4.1e+04	4.5e+03 /3.3e+04	4.2e+06/ 1.9e+06
autorefineMaxLevel	F	2.3e+08 /1.2e+09	4.9e+08 /1.3e+09	2.9e+05 /2.1e+06	3.2e+02 /2.4e+03	1.5e+03 /4.0e+04	3.0e+02 /1.3e+03	4.2e+06/ 7.4e+05
compressMass	F	2.3e+08 /1.2e+09	4.7e+08 /1.2e+09	3.0e+05 /1.3e+06	3.3e+02 /3.2e+03	1.6e+03 /3.0e+04	3.0e+02 /1.2e+03	4.1e+06/ 4.3e+05
diffPW	M	3.4e+08 /1.7e+09	8.1e+08 /3.3e+09	3.8e+05 /5.3e+06	1.2e+05 /2.6e+05	5.2e+03 /4.7e+04	5.5e+03 /3.8e+04	4.0e+06/ 1.8e+06
lbDeuxLoadProxy	F	2.4e+08 /1.2e+09	5.3e+08 /1.3e+09	1.6e+05 /2.6e+06	3.1e+02 /5.1e+02	1.7e+03 /4.1e+04	2.7e+02 /4.3e+02	4.1e+06/ 1.0e+06
norm2Estimate	F	2.9e+08 /1.0e+09	5.1e+08 /1.3e+09	1.2e+06 /3.0e+06	4.5e+02 /1.8e+03	2.1e+04 /5.4e+04	3.4e+02 /1.4e+03	3.8e+06/ 9.6e+05
pmapCutHistogram	F	2.8e+08 /1.2e+09	5.0e+08 /1.3e+09	7.5e+05 /2.0e+06	4.0e+02 /2.9e+03	9.8e+03 /3.7e+04	3.7e+02 /7.9e+02	4.0e+06/ 7.3e+05
truncateTolViolations	F	2.4e+08 /1.0e+09	4.4e+08 /1.3e+09	3.1e+05 /1.3e+06	3.5e+02 /2.1e+03	3.8e+03 /2.7e+04	3.0e+02 /6.0e+02	4.1e+06/ 4.2e+05
Total		2.2e+09/1.0e+10	4.5e+09/1.4e+10	3.8e+06/2.3e+07	2.4e+05/5.9e+05	5.1e+04/3.2e+05	1.2e+04/7.7e+04	3.3e+07/8.0e+06

Table 36: Per-pass performance for `TernaryTree`, ADT has 5 fields, SoA uses 3 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across -iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
add 1 tree	M	5/5	0%	0.0835	0.1018	0.0915	0.3450	0.91 $\times$	1.22$\times$	1.11$\times$	0.30 $\times$
sum tree	F	5/5	0%	0.0288	0.0505	0.0256	0.2733	1.13$\times$	1.75$\times$	1.97$\times$	0.18 $\times$
Total				0.1124	0.1523	0.1171	0.6183	0.96 $\times$	1.36$\times$	1.30$\times$	0.25 $\times$
Geomean								1.01 $\times$	1.46$\times$	1.48$\times$	0.23 $\times$

Table 37: Per-pass PAPI counters for `TernaryTree` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
add 1 tree	M	2.7e+08/ 2.3e+08	7.5e+08 /9.8e+08	4.1e+05 /1.1e+06	6.7e+04 /1.2e+05	3.4e+03 /1.1e+04	3.0e+03/ 2.7e+03	1.5e+06/ 1.1e+06
sum tree	F	1.5e+08/ 1.3e+08	4.6e+08 /5.4e+08	4.7e+04 /3.1e+05	1.1e+02/ 1.1e+02	1.2e+03 /9.0e+03	9.6e+01/ 9.3e+01	1.8e+06/ 1.3e+06
Total		4.2e+08/3.6e+08	1.2e+09/1.5e+09	4.6e+05/1.4e+06	6.7e+04/1.2e+05	4.6e+03/2.0e+04	3.1e+03/2.8e+03	3.3e+06/2.3e+06

Table 38: Per-pass PAPI counters for `TernaryTree` (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
add 1 tree	M	3.4e+08 /1.6e+09	9.5e+08 /2.4e+09	2.0e+05 /2.5e+06	7.5e+04 /1.8e+05	3.7e+03 /4.7e+04	2.9e+03 /4.8e+03	1.5e+06/ 9.4e+05
sum tree	F	2.6e+08 /1.4e+09	5.8e+08 /1.0e+09	8.1e+04 /2.4e+06	2.1e+02 /1.4e+03	1.4e+03 /4.2e+04	1.7e+02 /5.9e+02	1.4e+06/ 9.5e+05
Total		5.9e+08/3.0e+09	1.5e+09/3.4e+09	2.8e+05/4.9e+06	7.5e+04/1.8e+05	5.1e+03/8.9e+04	3.0e+03/5.4e+03	2.8e+06/1.9e+06

Table 39: Per-pass performance for `Trie`, ADT has 10 fields, SoA uses 9 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across -iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
autocompleteTopKProxy	F	5/10	50%	0.0146	0.0283	0.0115	0.1075	1.27$\times$	1.94$\times$	2.45$\times$	0.26 $\times$
countLazyNodes	F	4/10	60%	0.0125	0.0287	0.0075	0.1090	1.67$\times$	2.29$\times$	3.82$\times$	0.26 $\times$
countTerminals	F	3/10	70%	0.0120	0.0335	0.0145	0.1031	0.83 $\times$	2.80$\times$	2.31$\times$	0.32 $\times$
decayTrieStats	M	10/10	0%	0.0757	0.0815	0.0853	0.2458	0.89 $\times$	1.08 $\times$	0.96 $\times$	0.33 $\times$
resetTraversalState	M	10/10	0%	0.0751	0.0823	0.0812	0.2338	0.92 $\times$	1.10 $\times$	1.01 $\times$	0.35 $\times$
sumPrefixFreq	F	3/10	70%	0.0122	0.0481	0.0177	0.1167	0.69 $\times$	3.95$\times$	2.71$\times$	0.41 $\times$
sumSubtreeHints	F	3/10	70%	0.0119	0.0311	0.0115	0.1060	1.03 $\times$	2.61$\times$	2.70$\times$	0.29 $\times$
Total				0.2140	0.3334	0.2293	1.022	0.93 $\times$	1.56$\times$	1.45$\times$	0.33 $\times$
Geomean								1.00 $\times$	2.05$\times$	2.05$\times$	0.32 $\times$

Table 40: Per-pass PAPI counters for `Trie` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
autocompleteTopKProxy	F	7.4e+07/ 3.9e+07	1.3e+08 /2.1e+08	1.5e+06/ 6.6e+05	1.7e+02/ 1.2e+02	5.3e+04/ 9.3e+03	1.6e+02/ 1.1e+02	3.8e+06/ 6.0e+05
countLazyNodes	F	6.3e+07/ 3.2e+07	1.3e+08 /1.6e+08	1.2e+06/ 2.9e+04	2.2e+02 /2.5e+02	1.6e+04/ 3.3e+03	2.0e+02 /2.4e+02	4.1e+06/ 6.2e+05
countTerminals	F	6.0e+07/ 3.0e+07	1.0e+08 /1.2e+08	1.3e+06/ 6.4e+05	3.4e+02/ 3.0e+02	2.5e+04/ 1.3e+04	3.2e+02/ 2.6e+02	4.1e+06/ 2.0e+05
decayTrieStats	M	1.6e+08 /2.1e+08	3.5e+08 /7.9e+08	7.3e+05 /1.2e+06	9.1e+04 /1.1e+05	5.6e+03 /1.9e+04	2.7e+03 /2.2e+04	2.9e+06/ 1.2e+06
resetTraversalState	M	1.6e+08 /2.0e+08	2.8e+08 /7.0e+08	1.0e+06/ 7.7e+05	8.9e+04 /1.1e+05	7.8e+03 /1.2e+04	2.6e+03 /9.7e+03	3.3e+06/ 9.0e+05
sumPrefixFreq	F	6.1e+07/ 3.0e+07	1.1e+08 /1.2e+08	1.3e+06/ 6.4e+05	5.4e+02/ 5.0e+02	2.8e+04/ 1.3e+04	4.1e+02/ 3.8e+02	4.1e+06/ 1.9e+05
sumSubtreeHints	F	6.0e+07/ 3.0e+07	1.1e+08 /1.2e+08	1.4e+06/ 6.4e+05	3.4e+02/ 2.2e+02	2.7e+04/ 1.3e+04	3.2e+02/ 2.1e+02	4.0e+06/ 2.4e+05
Total		6.5e+08/5.7e+08	1.2e+09/2.2e+09	8.4e+06/4.6e+06	1.8e+05/2.2e+05	1.6e+05/8.2e+04	6.6e+03/3.3e+04	2.6e+07/3.9e+06

Table 41: Per-pass PAPI counters for `Trie` (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
autocompleteTopKProxy	F	1.4e+08 /5.5e+08	2.3e+08 /7.5e+08	1.1e+06 /1.8e+06	2.2e+02 /1.0e+03	3.2e+03 /3.4e+04	2.0e+02 /7.5e+02	2.8e+06/ 4.5e+05
countLazyNodes	F	1.5e+08 /5.5e+08	2.2e+08 /7.6e+08	2.2e+06/ 1.2e+06	1.1e+02 /1.2e+03	2.7e+03 /2.5e+04	9.5e+01 /6.1e+02	2.7e+06/ 3.3e+05
countTerminals	F	1.7e+08 /5.2e+08	2.0e+08 /7.3e+08	4.1e+06/ 6.9e+05	1.8e+02 /1.2e+03	1.0e+05/ 1.7e+04	1.6e+02 /3.2e+02	2.5e+06/ 1.9e+05
decayTrieStats	M	1.9e+08 /1.0e+09	4.5e+08 /2.1e+09	3.1e+05 /5.2e+06	9.1e+04 /3.7e+05	3.5e+03 /3.6e+04	3.6e+03 /3.4e+04	2.9e+06/ 1.2e+06
resetTraversalState	M	2.0e+08 /9.6e+08	3.7e+08 /2.0e+09	2.8e+05 /3.3e+06	1.1e+05 /1.5e+05	4.9e+03 /2.9e+04	3.3e+03 /3.5e+04	3.0e+06/ 9.6e+05
sumPrefixFreq	F	1.5e+08 /5.4e+08	2.1e+08 /7.3e+08	4.2e+06/ 7.0e+05	5.4e+02 /1.3e+03	8.8e+04/ 1.7e+04	4.2e+02 /4.4e+02	2.3e+06/ 1.9e+05
sumSubtreeHints	F	1.6e+08 /5.4e+08	2.1e+08 /7.3e+08	3.0e+06/ 6.9e+05	1.4e+02 /9.4e+02	5.4e+04/ 1.7e+04	1.3e+02 /2.0e+02	2.6e+06/ 1.9e+05
Total		1.2e+09/4.7e+09	1.9e+09/7.8e+09	1.5e+07/1.4e+07	2.0e+05/5.3e+05	2.6e+05/1.7e+05	7.9e+03/7.2e+04	1.9e+07/3.5e+06

Table 42: Per-pass performance for `ColorOmtree`, ADT has 25 fields, SoA uses 18 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across -iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
paletteEntriesQuantized	F	13/25	48%	0.0439	0.0556	0.0364	<i>OOM</i>	1.20 $\times$	1.27 $\times$	1.53 $\times$	–
quantizationErrorProxy	F	10/25	60%	0.0462	0.0663	0.0430	<i>OOM</i>	1.08 $\times$	1.44 $\times$	1.54 $\times$	–
Total				0.0901	0.1219	0.0794	–	1.13 $\times$	1.35 $\times$	1.54 $\times$	–
Geomean								1.14 $\times$	1.35 $\times$	1.54 $\times$	–

Table 43: Per-pass PAPI counters for `ColorOmtree` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
paletteEntriesQuantized	F	2.1e+08/ 1.8e+08	6.7e+08 /7.6e+08	5.7e+06/ 1.5e+06	5.4e+02/ 5.3e+02	4.5e+04 /4.7e+04	3.6e+02 /3.8e+02	9.4e+06/ 1.4e+06
quantizationErrorProxy	F	2.3e+08/ 2.2e+08	9.7e+08 /1.2e+09	4.2e+06/ 8.9e+05	1.9e+02 /2.9e+02	1.6e+04 /3.8e+04	1.5e+02 /2.7e+02	9.4e+06/ 2.9e+06
Total		4.4e+08/4.0e+08	1.6e+09/2.0e+09	9.9e+06/2.4e+06	7.3e+02/8.2e+02	6.1e+04/8.5e+04	5.2e+02/6.5e+02	1.9e+07/4.3e+06

Table 44: Per-pass PAPI counters for `ColorOmtree` (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
paletteEntriesQuantized	F	2.8e+08/–	7.8e+08/–	2.1e+06/–	7.3e+02/–	4.5e+03/–	5.9e+02/–	8.4e+06/–
quantizationErrorProxy	F	3.4e+08/–	1.1e+09/–	1.5e+06/–	6.1e+02/–	2.3e+04/–	6.0e+02/–	7.2e+06/–
Total		6.2e+08/–	1.8e+09/–	3.6e+06/–	1.3e+03/–	2.8e+04/–	1.2e+03/–	1.6e+07/–